

RUVARASHE SHALOM MADZIME

Cape Town, South Africa | 4158028@myuwc.ac.za | [linkedin.com/in/ruvarashe-madzime](https://www.linkedin.com/in/ruvarashe-madzime) | github.com/rue404NotFound

RESEARCH INTERESTS

Formal non-monotonic reasoning, in particular the KLM framework, and its use for giving learned rule sets a logical reading. Moving toward neurosymbolic AI, focused on how logical structure can be embedded into neural networks during training and how networks might learn symbolic representations directly from data.

EDUCATION

MSc Computer Science, University of the Western Cape and CAIR *Feb 2025 to Nov 2026*

Cape Town, South Africa

- Thesis: *Toward a Defeasible Semantics for Symbolic Classifiers*. Supervisors: Prof. Thomas Meyer, Prof. Louise Leenen.
- Area: knowledge representation, defeasible reasoning, interpretable machine learning.

BSc (Hons) Computer Science, cum laude (85%), University of the Western Cape *Jan 2024 to Nov 2024*

Cape Town, South Africa

- Modules: artificial intelligence, applied data science, IoT, cybersecurity, big data engineering.

BSc Computer Science, cum laude (84%), University of the Western Cape *Mar 2021 to Nov 2023*

Cape Town, South Africa

- Key modules: artificial intelligence, mathematics, statistics.

PUBLICATIONS

- Madzime, R., and Meyer, T. (2026). Toward Defeasible Machine Learning: When Learned Rule Sets Become Defeasible Theories. In *Proceedings of the 24th International Workshop on Nonmonotonic Reasoning (NMR 2026)*, Federated Logic Conference (FLoC), Lisbon, Portugal.
- Karge, J., Ferguson, R., Grimaldi, D., Haldimann, J., Madzime, R., and Meyer, T. (2026). First Steps Towards Human-AI Ranking Aggregation. In *Proceedings of the Joint Workshop on Statistics and Knowledge Integration for Logic, Learning, Ethical Decisions, and LLMs (SKILLED-LLMs 2026)*, Federated Logic Conference (FLoC), Lisbon, Portugal.
- Madzime, R. S., Meyer, T., and Leenen, L. (2025). An Override-Aware Classifier for Transparent AI. In *Proceedings of the 6th Southern African Conference on Artificial Intelligence Research (SACAIR 2025)*, Cape Town, South Africa.
- Madzime, R., and Nyirenda, C. (2024). Enhanced Electronic Health Record Text Summarisation Using Large Language Models. In *Proceedings of the 5th Southern African Conference on Artificial Intelligence Research (SACAIR 2024)*, Bloemfontein, South Africa.

EXPERIENCE

Research Assistant (SARCHI NRF), University of the Western Cape *Mar 2026 to present*

Cape Town, South Africa

- Contribute to research in knowledge representation as part of the SARCHI programme under Prof. Louise Leenen.
- Teaching support for undergraduate and postgraduate CS modules; co-supervise Honours research projects.

Graduate Lecturing Assistant, University of the Western Cape *Feb 2025 to Nov 2025*

Cape Town, South Africa

- Ran tutorials and consultations for CS modules, supporting over 50 students.
- Graded assignments and exams; supported lecturers with coursework design.

Graduate Research Assistant, University of the Western Cape *Jan 2024 to Nov 2024*

Cape Town, South Africa

- Built an LLM-based summarisation system for Electronic Health Records, published at SACAIR 2024.
- Designed the full pipeline from dataset analysis through model evaluation in Python.

Python Software Developer, Apparel Technology Inc. *Oct 2023 to Mar 2024*

Cape Town, South Africa

- Built a Python and Arduino monitoring system that tracked factory machine uptime in real time.
- Created a data visualisation dashboard with Python Dash for the operations team.

HONOURS AND AWARDS

- UWC Mastercard Foundation Scholarship, 2025 to 2026.
- Top Computer Science Student, University of the Western Cape, 2023.
- KnowBe4 Women in Cybersecurity Scholarship, 2023.
- Dean's Merit Award, University of the Western Cape, 2021, 2022, 2023, and 2024.

TECHNICAL SKILLS

Languages and tools: Python, Java, SQL, Answer Set Programming (Clingo), SMT solvers (Z3), Git, LaTeX.

AI and research: PyTorch, scikit-learn, Hugging Face, SHAP, defeasible reasoning, knowledge representation, NLP, LLMs, retrieval-augmented generation, cloud deployment.